

1. Define HDL language:

HDL stands for Hardware Description Language. It is used to describe the structure and behavior of digital electronic circuits. HDL allows simulation, design, and synthesis of digital systems like microprocessors, memory chips, and FPGAs. The two most common HDLs are Verilog and VHDL.

2. What do you mean by Verilog Module, Module Declaration and Architecture Body?

- Verilog Module:

A Verilog module is the basic building block of any Verilog design. It represents a hardware component or circuit and contains the logic to describe its behavior.

- Module Declaration:

The module declaration is the starting point of a Verilog module. It defines the name of the module, its input and output ports.

Example:

```
module AND_gate (input A, input B, output Y);
```

- Architecture Body:

The architecture body (or module body) contains the internal working of the module – how outputs are generated from inputs.

Example:

```
assign Y = A & B;  
endmodule
```

3. What do you mean by Data Type used in Verilog?

In Verilog, data types define the kind of signals or variables used in a design.

Common data types:

- wire: used for connections (combinational circuits)

- reg: used to store values (sequential logic)

- integer: used in testbenches or for loops

- real: used for real number values (simulation only)

These data types help the compiler understand how the hardware will behave.

4. What is the difference between Gate Level Modeling and Dataflow Modeling?

Feature	Gate Level Modeling	Dataflow Modeling
Description	Describes circuit using gates	Using boolean expressions
Style	Low-level	High-level
Keywords	and, or, not, xor	assign
Example	and (Y, A, B);	assign Y = A & B;
Use	Detailed gate connections	Clear logic expression